Design Journal

Xianglong Lin

45791439

Design Computing Studio 2 – Interaction Design (DECO2850)

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Design Journal

Student Name: Xianglong Lin Student Number: 45791439 Team Name: H2OMG

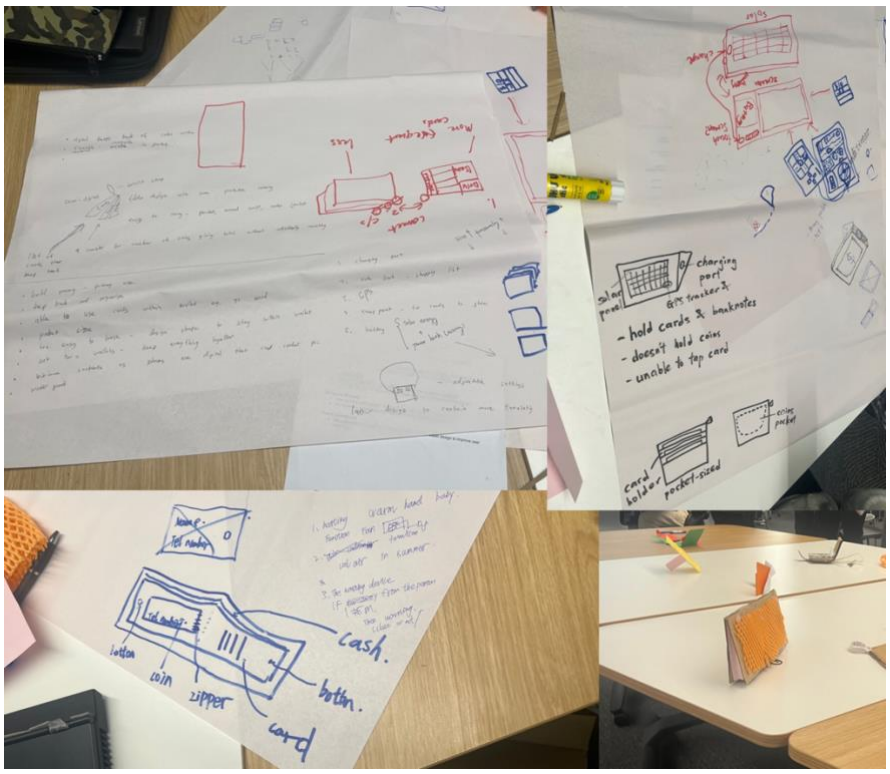
Week: _1_

Title: Wallet Design

Description:

In the 1st week, our team was assigned to participate in a design session. The team we formed began brainstorming and designing an ideal wallet. The original intention of the activity was to exercise our design thinking and teamwork skills and spirit. The activity content involved designing 5 different wallet designs, interviewing team members to understand their experience with current wallets on the market, providing feedback and reflection on the design sketches, and comprehensively improving the final wallet design scheme.

Visual Evidence:



Statement on Learning:

This week's activities repeatedly emphasised the importance and necessity of empathy in design. The full implementation of research and interviews revealed the main pain points of users, such as the common problem of difficult placement of cards and the need for advanced features such as chargers. The design of multiple sketches actively promoted us to seek different solutions and find a balance between functionality and aesthetics. The feedback stage improved our team's understanding of user preferences. My experience and conclusion is that the function of a wallet is directly proportional to its size, and an increase in function is likely to lead to an increase in size, and vice versa. I was impressed by the wallet designs of the team members and other students, such as those that can print money, those that are folding, and those that use interlocking designs to expand functional areas. On the balance, this week's learning and practice opened my eyes to the depth of interactive design, and also made me realise the importance of teamwork communication and user feedback in design. Next, I plan to apply the knowledge and insights I gained from this week's studio experience to our upcoming projects and start building a team, striving to integrate and apply knowledge in a flexible and comprehensive way.

Design Computing Studio 2

Design Journal

Student Name: Xianglong Lin Student Number: 45791439 Team Name: H2OMG

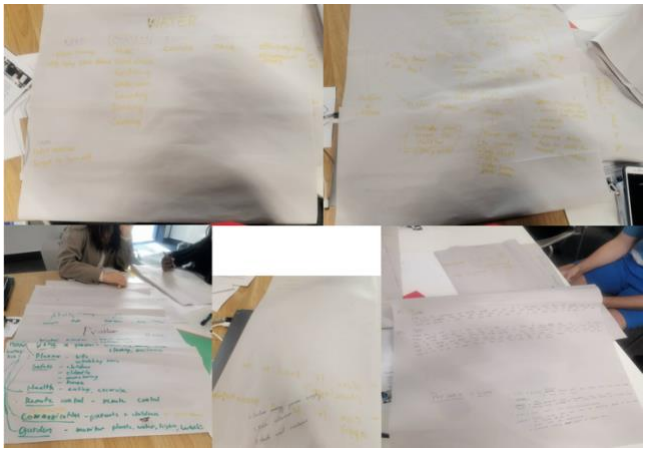
Week: _2_

Title: Team Formation and Project Focus

Description:

In the 2nd week, I proactively collaborated with some friends from the previous week's wallet design to form a project team under the guidance of our tutors. We successfully formed a team with 2 additional members, each bringing unique educational backgrounds, skills, and experiences. After getting to know each other, we brainstormed around 3 themes: Home & Domestic Setting, Office & Workplace Settings, and Health & Wellbeing to decide our future direction. I preferred Theme 2, Office & Workplace Settings, due to my background as an IT project manager at Huahong International Law Firm. However, we ultimately agreed on Theme 1, Home & Domestic Setting, as it resonated more with each team member's personal experiences and insights. We then narrowed Theme 1 down to a more specific focus: communication among household members or water usage. Upon further brainstorming and research, we discovered that a project centred on water usage was more feasible compared to communication. Water is a universal need, and everyone requires it similarly, whereas communication methods vary greatly from household to household and depend on individual circumstances. After selecting our main theme and specific focus, we created two personas, Trump and Jeff, and developed corresponding scenarios. We also established a WhatsApp group to facilitate ongoing communication after the studio session.

Visual Evidence:



Statement on Learning:

The most important aspect of this studio was the core design principles emphasised by the tutor: 1. Who is the user? 2. What does the design do? 3. How does the design achieve this? Additionally, the common concern of whether the final solution should be an app, or a website was addressed, highlighting that at this initial stage, we should not focus on technology or worry about the final form of the solution. Adhering to these core principles, the team formation allowed us to break down the theme, making it more manageable for the project. We structured and divided the tasks, assigning each person roles based on their skills and strengths. The creation of Personas focused on the users' water usage needs, prioritising water recycling and minimising waste. This not only helped us clearly define the project goals and direction but also ensured the practicality and relevance of our design solution. Throughout this process, I learned how detailed user persona analysis can guide design decisions and enhance team collaboration, ultimately leading to more effective problem-solving.

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Student Name: Xianglong Lin Student Number: 45791439 Team Name: H2OMG

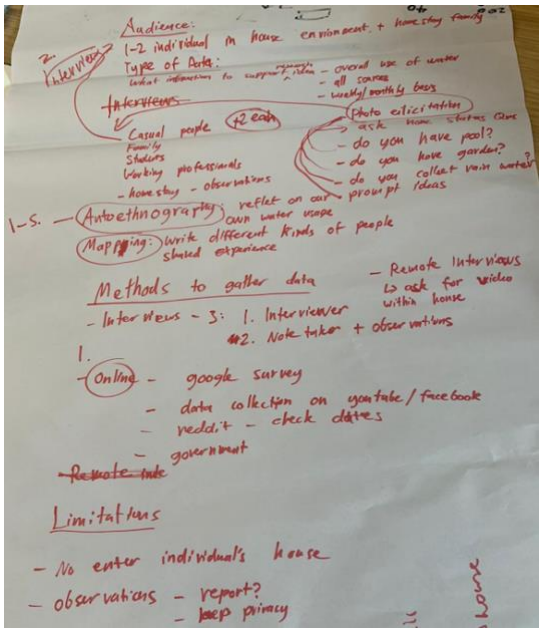
Week: _3_

Title: Method Selection in Water Resource Design

Description:

In the 3rd week, our team moved from initial brainstorming to a more focused exploration of water resource management within households. Building on the personas established in the 2nd week, we decided to use Interviews and Autoethnography as our primary research methods. These 2 approaches allow us to study water usage from both external and internal perspectives, providing a comprehensive view. Last week, we explored broader themes like household communication and water usage. However, we realised the need to focus specifically on water resource usage to develop measurable and feasible design solutions. Unlike the previous week, which centred on defining the project and personas, this week we concentrated on selecting research methods that best uncover user behaviours and personal experiences related to water usage. Interviews will help us gather insights into the social and cultural contexts influencing water usage, while Autoethnography will provide deeper, introspective reflections, complementing the data from interviews. This combination ensures a thorough approach to our project.

Visual Evidence:



Statement on Learning:

This week's studio taught me the importance of balancing different research methods to better capture user behaviors and experiences. Through Interviews, a wide range of perspectives will be captured, ensuring that the design solutions are grounded in the real experiences of different user groups. Autoethnography allowed me to critically reflect on my own interactions with water usage, bringing to light subconscious habits and preferences that might otherwise go unnoticed. The combination of these methods taught me that integrating external data collection with personal self-reflection can lead to more comprehensive and user-centred design outcomes. In summary, the method selection and planning this week undoubtedly lay the foundation for our team's upcoming research activities. By leveraging both in-depth and accessible research methods, we will gain a more thorough understanding of the complexities of water resource usage within household environments, ultimately ensuring that our design solutions are effective and user-centred.

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Student Name: Xianglong Lin Student Number: 45791439 Team Name: H2OMG

Week: _4_

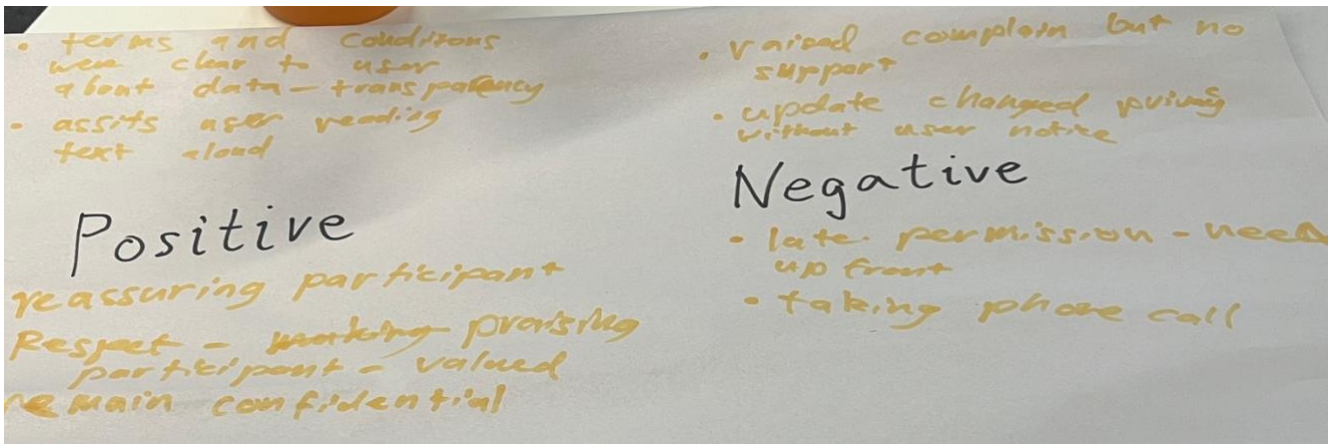
Title: Ethical Implications in Water Usage Tracking

Description:

In Week 4, our team's attention was drawn to ethical considerations in the design project, particularly concerning the tracking of water usage in households. Building on the research methods established in Week 3, this week we focused on addressing potential user privacy issues that may arise during the collection of water consumption data. We conducted an in-depth analysis of a scenario involving a father who is surprised and frustrated by a high water bill, struggling to identify the cause of the excessive water usage. Despite his efforts to reduce water consumption, he is still unable to pinpoint the problem, and therefore a more efficient solution is needed. This illustrative scenario helped us recognise several key ethical issues:

1. Confidentiality: Water usage data should be securely stored and accessible only to the user, preventing unauthorised access and privacy breaches.
2. Consent for Monitoring: Users must be informed and fully aware of what data will be collected, how it will be used, and who will have access to it. This requires us to establish clear and transparent communication upfront, covering the purpose of monitoring and ensuring the integrity of voluntary participation.
3. Private Spaces Within the Home: Our design must not intrude on areas that users consider private, such as bedrooms or bathrooms, and in any case, it is safest to seek permission. This is an important measure to maintain appropriate boundaries and avoid invading privacy.

Visual Evidence:



Statement on Learning:

This week's studio on ethical considerations has made me fully aware of the importance and necessity of considering user privacy and consent when designing project solutions. Through the analysis of the specific water usage scenario, I concluded that ethical design is not only about avoiding harm but also about respecting the user's autonomy and personal space. Additionally, I have gained a renewed understanding of the concept of minimising harm in design. Although data is an indispensable part of designing solutions, it is essential to ensure that the user is informed and comfortable with the way data is collected. This principle will undoubtedly be fundamental to the success of our final design in solving the problem while adhering to ethical standards. Overall, careful consideration of ethical issues and the formulation of guidelines will lead our team to create solutions that are both innovative and responsible, truly achieving a user-centred approach!

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Student Name: Xianglong Lin Student Number: 45791439 Team Name: H2OMG

Week: _5_

Title: User Needs, Problems and Water-Saving Design Solutions

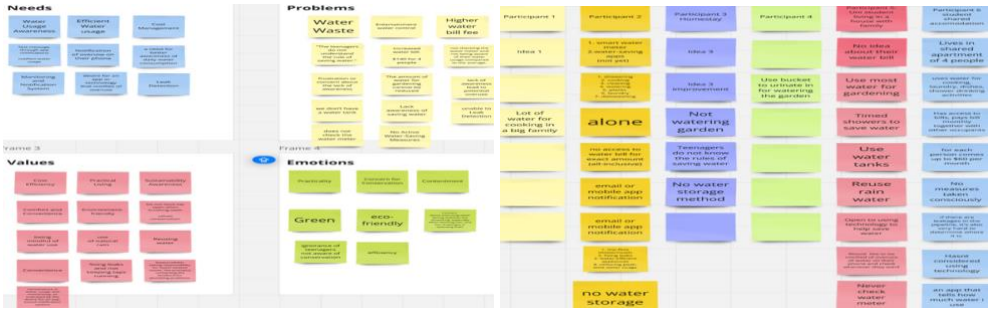
Description:

In the 5th week, our team analysed the interview data collected on tracking water usage immediately after implementing ethical considerations in the 4th week. We tried to use affinity diagrams to organise and fully understand the collected data, which visualises and categorises relevant qualitative data that is key to influencing design decisions and solutions.

The affinity diagram analysis reveals 4 important insights:

1. Needs: Most users have a strong need for water awareness, and therefore desire real-time monitoring and notifications to better manage their water consumption. On the other hand, efficient water use and cost management are also valued. Users hope that there is a tool on the market to detect and inform of leaks and excessive usage.
2. Problems: The issue of water waste is one of the more prominent problems, based on a lack of awareness and measures to save water. Users are usually frustrated by the increased water bill but also worried about how to track individual usage in a multi-person living environment.
3. Values: Users are focused on sustainability awareness and cost-effectiveness, preferring more environmentally friendly solutions without compromising convenience or sacrificing practicality. Meanwhile, comfort and convenience are also particularly important. The system is expected to seamlessly integrate with users' daily lives while minimising disruption of traditional habits.
4. Emotions: The data covers a complex mix of emotions, including concern for resource conservation, practicality, and strong environmental awareness. This is manifested in users feeling satisfied and proud when contributing to resource conservation, but also experiencing frustration due to a lack of efficient tools and relevant knowledge.

Visual Evidence:



Statement on Learning:

This studio used affinity diagram analysis data to demonstrate the importance of organising qualitative data into meaningful categories. User needs, problems, values, and emotions reveal the design areas of solutions while helping us understand data. One learning experience is that organised data analysis is good at being used to discover user pain points that are not easily detected. For example, the waste of water resources is often closely related to users' lack of awareness and measures for saving water. Therefore, simply developing monitoring tools is not enough; instead, solutions that provide feasible suggestions are needed. For example, the system can propose specific measures to reduce water consumption based on users' habits or patterns of using water. For instance, it can suggest shortening the time for watering gardens and reducing unnecessary flushing to lower water bills.

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Student Name: Xianglong Lin Student Number: 45791439 Team Name: H2OMG

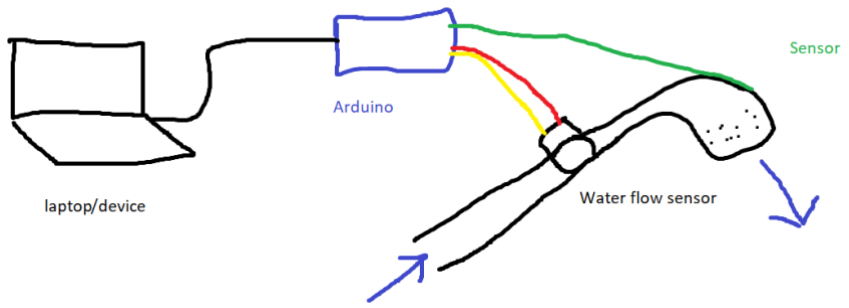
Week: _6_

Title: Prototyping with Arduino for Water Flow Monitoring

Description:

In the 6th week, our team moved to the implementation phase of prototyping based on user needs and issues identified from last week. To meet the demand of users for efficient monitoring tools, a combination solution of physical components and digital technology was explored. Initially, we were all hindered by our lack of experience in Arduino and backend programming, which we tried to avoid and bypass. However, after constant communication and consultation with the tutors, we gradually realised that a purely digital solution was not feasible. Therefore, we had initially developed a prototype that mainly consists of an Arduino Uno and a water flow sensor to obtain water flow data for further suggestions. Although we know that there will be technical and programming difficulties, we are confident that with the help of the tutors, we can overcome them and ultimately achieve a more comprehensive portfolio solution.

Visual Evidence:



Statement on Learning:

This studio had pushed us from the theoretical stage of data analysis last week to the next practical implementation stage. The final adoption of Arduino for the project design was a significant step in overcoming our psychological fears and stepping out of our comfort zone. This demonstrated each team member's drive to improve, the willingness to face challenges head-on, and the teamwork spirit that is not afraid of difficulties. Although we only focused on identifying the need for real-time monitoring tools for water resources last week, it seems that converting conceptual stage outputs into operational models at the prototype stage is more complex than expected. This involves a range of uncertain factors such as the technical background of the team and material procurement. Furthermore, the innovative approach of integrating digital solutions with physical hardware had provided us with new insights and once again solidified our adaptability to solution design and teamwork skills. In my case, I initially proposed considerations for front-end development due to my area of expertise. However, integrating Arduino hardware quickly made me realise the importance of understanding the entire system in order to successfully design a final prototype. Although this growth will most likely make me more inevitably exposed to the backend development aspects that I previously feared, it will undoubtedly increase my backend programming skills and experience. Meanwhile, the iteration of prototype design had also begun to show, it is more practical to pursue "good enough" rather than perfection in the early stages of design. This set of learning and insight growth made the progress in our team's user-centeredness and technical feasibility.

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Student Name: Xianglong Lin Student Number: 45791439 Team Name: H2OMG

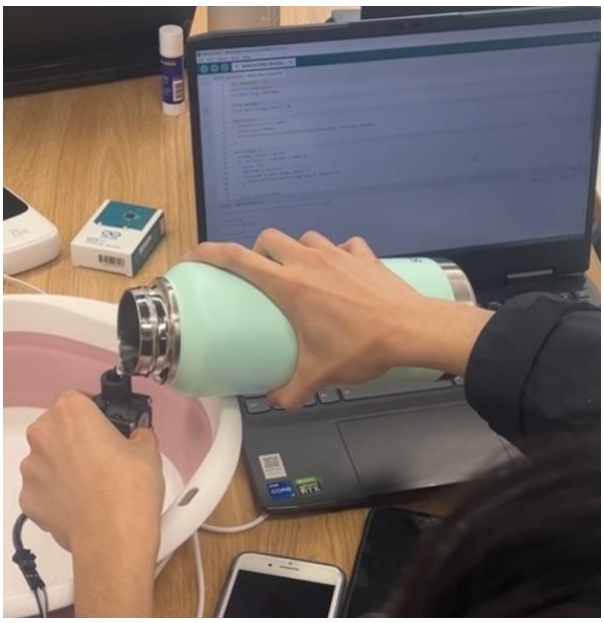
Week: _7_

Title: Design Exhibit of Iteration 1

Description:

In the 7th week of Design Exhibit of Iteration 1, our team worked together to present our water resource project profile, prototype design map, interview transcripts, affinity diagram, concept, storyboards, low-fidelity prototype, and hardware dynamic of physical interactions. I mainly explained Storyboard 1 & 2, that explained how our system helps target users solve the problem of water waste. In Storyboard 1, I described how a father used our system to discover excessive water usage in a malfunctioning sprinkler system in the garden. While Storyboard 2 demonstrated how a family could use our system to monitor the duration of shower water usage and obtain relevant solution suggestions through the app. This first Design Exhibit not only presented our specific interaction methods of hardware and real-time water monitoring, but also provided valuable feedback and suggestions from the tutors on how to improve dynamic functions and optimise the user interface in the next iteration stage.

Visual Evidence:



Statement on Learning:

The storyboards of this Design Exhibit made me deeply understand that it is necessary and important but also challenging to transform complex and professional digital technology solutions into a target form that is easy for target users to understand and feasible to operate. Although our Arduino had now achieved the ability to detect and calculate the amount of water flowing through the device within 10 seconds, it is worth considering how to convey this data in a form that users prefer rather than simply presenting it as data at this stage. For example, this requires us to try to design and create app notifications and reminder functions that are based on the data values of water flow, such as shortening shower time and repairing water leaks mentioned in the storyboards. Meanwhile, this made us realise the importance of simplifying technical explanations and how to operate and implement them, which would be particularly helpful for non-technical users or edge users. In the next iteration, we will develop the low-fidelity prototype into a high-fidelity prototype and start developing the front-end pages in succession. We will also conduct user tests to improve the clarity and usability of our water resource project as expected by users.

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Student Name: Xianglong Lin Student Number: 45791439 Team Name: H2OMG

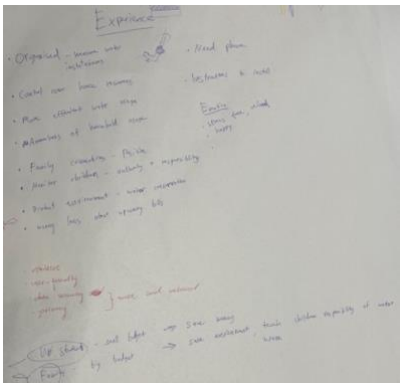
Week: _8_

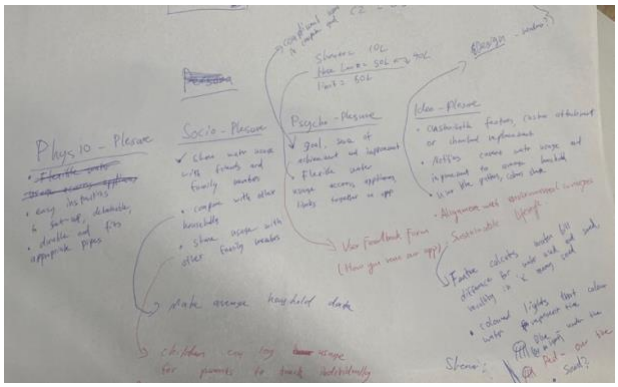
Title: Balancing Utility and Enjoyment in Water Conservation through the Four Pleasures

Description:

In the 8th week, I began to try to incorporate user experience (UX) into prototype design based on the prompts in the studio material. According to the feedback from the tutors of last week's Design Exhibit of Iteration 1, the development of the front-end interface had been started to achieve user interaction and data visualisation for water resource projects. Although the Arduino-based water flow sensor is effective in collecting and presenting raw data, a more user-friendly and actionable form is even more important. Therefore, while conceiving the layout and interaction flow of the high-fidelity prototype, I consciously focused on application notifications and customised water-saving suggestions based on real-time data. In addition, I also applied Patrick Jordan's Four Pleasures to the project during the studio in order to ensure that the functionality of the prototype would be practical, and the user interface would be intuitive and effective.

Visual Evidence:





Statement on Learning:

This studio focused on improving user experience through usability and emotional engagement. Based on the feedback from Exhibit 1, although my prototype demonstrated satisfied performance on the technical level, users might have difficulty understanding the meaning of the data in an effective and intuitive form. Therefore, it was necessary to adopt a user-centred design approach at this moment, especially in terms of clarity, intuition, and operability of the user interface. As I iterate from low-fidelity prototype to high-fidelity prototype, the front-end prioritises ensuring that users visually receive reasonable and meaningful information notifications instead of simply displaying data analysis charts. For example, the app proposes specific suggestions for shortening water usage time or changing water usage habits based on water volume values, which will also directly support users' goal of saving water.

On the other hand, the analysis of Four Pleasures on this project led me to further optimise and iterate the design from an emotional and social perspective. In terms of physio-pleasure, I design and create visually intuitive and user-friendly front-end interfaces for a more positive sensory experience; In terms of socio-pleasure, I plan to enable users to share water-saving achievements with friends and family, promoting social experience and enhancing collective pride and responsibility for sustainability; In terms of psycho-pleasure, users not only get clear and effective suggestions from the app on water conservation, but also gain a sense of accomplishment when approaching and achieving water conservation goals; In terms of ideo-pleasure, the system concept of the app is highly consistent with the user's aim of saving water and protecting the environment, allowing users to feel that they have contributed their own strength to environmental protection.

This balanced approach had profound significance for both system functionality and user experience. Once the front-end interface is completed, more user tests will be conducted to improve these aspects and achieve the project's satisfaction with diverse needs. This also stood for the foundation of my user interface in meeting users' actual needs and evoking emotional resonances from data to user-friendly actionable insights.

Design Computing Studio 2

Design Journal

Student Name: Xianglong Lin Student Number: 45791439 Team Name: H2OMG

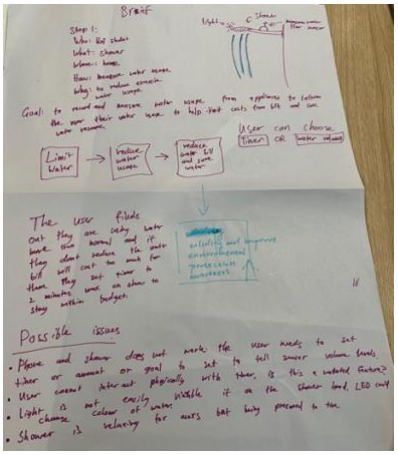
Week: _9_

Title: Balancing User Experience and Technical Performance through User Enactment

Description:

In the 9th week, I conducted user enactment in the studio to simulate the interaction between users and the water conservation system in real-life scenarios. According to the high-fidelity prototype of last week, I designed a scenario where a university student controlled the duration of water usage through the water system during the shower. The device utilised the water flow tracking feature enabled to users, allowing users to customise their water usage and receive real-time water usage reminders through the lights (Green: Normal; Yellow: Critical Value; Red: Overused) to limit usage and receive suggestions for reducing shower time. This user enactment vividly demonstrated the assessment of the intuitive nature of notifications and their effectiveness in guiding users to make real-time water-saving responses and actions. In addition, during the mid-semester break, I hosted the group meeting to determine the important guidelines for using HTML, CSS, and JavaScript as the front-end development programming language for the website.

Visual Evidence:



Statement on Learning:

The interesting team task of user enactment in this studio not only built user scenarios, but also highlighted the pertinence and importance of simulating user behaviour to improve the user experience, especially the management of water resources during the shower process, even though the equipment and system were suitable for various water use scenarios. Although the high-fidelity prototype last week was able to detect and provide real-time data well, this simulation revealed that there was still room for improvement in notifications to ensure the system was intuitive and user-friendly. For example, it was a bit challenging to ensure that the lights were placed in a prominent position to remind users about water usage while also avoiding contact with water as much as possible.

Unlike last week's focus on technical functions, this user enactment reflected the necessity and importance of user experience. The prompts and notifications should be simple and applicable, so that users could notice as soon as possible. For example, when the green light turns to yellow, users will know how long it will take for the water usage they have set to reach the limit. However, it is not practical to display a prompt pop-up only on devices such as users' mobile phones, as most people will avoid bringing their phones into the shower.

This user enactment emphasised the balance between technical performance and user experience. Real-time notifications assist users in making water-saving measures, but meanwhile, ensuring that the notifications are contextually appropriate and motivating is a top priority, so that the users' sense of accomplishment in approaching the established water-saving goals enhances the users' psycho-pleasure, as mentioned by last studio. This discovery also prompted us to prioritise developing web-based systems using HTML, CSS, and JavaScript, which are simple and easy to implement not only in terms of user interface but also in terms of programming and development difficulty, rather than building mobile applications.

Design Computing Studio 2

Design Journal

Student Name: Xianglong Lin Student Number: 45791439 Team Name: H2OMG

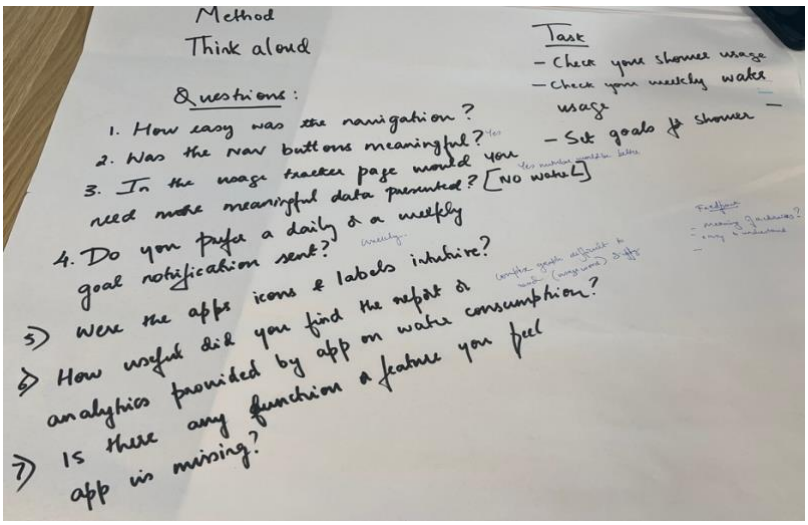
Week: _10_

Title: Enhancing User Usability through Think-Aloud Evaluation

Description:

In the 10th week, my focus in the studio was user evaluation, which was using think-aloud to evaluate the usability and user experience of the current high-fidelity prototype of the water conservation system. This protocol emphasised the user to express ideas and feelings about using and interacting with different functions during the operation of the water system. During the evaluation process, a representative from another team was sent to my team. The user's tasks such as setting water use goals and interpreting water usage analysis were observed and I asked the representative about the experience using them, such as “How useful is the water usage report?” and “Do you prefer daily or weekly goal notifications?” These questions helped me better understand user needs and preferences. This test provided valuable insights into the user experience of the prototype, such as the difficulties the user encountered when interpreting data and the need for more intuitive navigation buttons.

Visual Evidence:



Statement on Learning:

This week's studio activity highlights the critical role of direct interaction with users in order to collect real usability feedback. Compared to last week's user enactment activity, think-aloud enabled me to gain a deeper understanding of specific aspects and details of user interaction, such as the intuitive nature of navigation and the clarity of information presentation. For example, users had difficulty understanding certain icons and analysis charts, which had made me realise the importance of clearer labels and more concise presentation methods, both in prototype design and front-end interface coding. This feedback is different from the activity simulation of the controlled scenario last week, which was more user-oriented in a specific shower room setting.

The feedback from the think-aloud prompted significant adjustments to the next design iteration. Specifically, the user interface is needed to be enhanced, clearer navigation buttons should be used to make the analysis data more easily understood visually, and perhaps the complex information (e.g. water flow amount) should be simplified into a more intuitive presentation form. Unlike last week's user enactment, this week's activity leveraged real user experience to improve basic usability. Next, a portion of the front-end interface will be redesigned to make it more user-friendly and engaging, ensuring that users can interact with actionable information in a visually appealing way.

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Design Journal

Student Name: Xianglong Lin Student Number: 45791439 Team Name: H2OMG

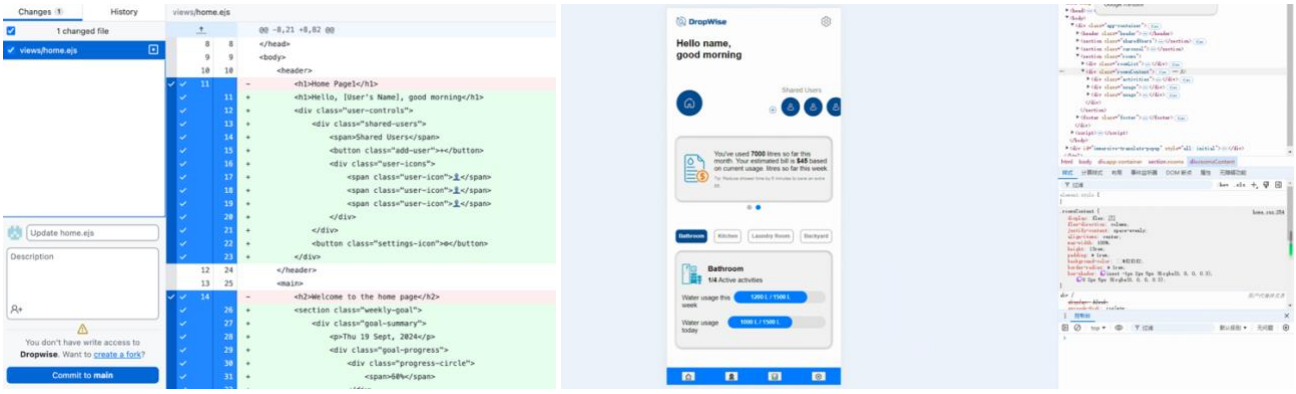
Week: _11_

Title: EJS Dynamic Front-End Page, GitHub Desktop Code Sharing, Privacy and Security Considerations

Description:

In the 11th week, the main task was to prepare for the upcoming Exhibit 2. I took the initiative to receive the task of front-end coding (EJS) for the "Home" and "Setting" pages of the water resource protection system. GitHub Desktop is used to share and synchronise front-end code, allowing other front-end team members to clearly see code differences and comments through automatic highlights. Although some team members using Windows systems had some connection issues when configuring local to GitHub due to a lack of experience, fortunately, I had practice in another DECO course this semester that helped solve the problem smoothly. In addition, I urged all dynamic pages to be created and integrated as soon as possible, preferably by the end of this week. This is to allow at least 1 week for the integration of front-end and back-end (Java) and the connection to the SQL user account database, and to allow for systematic user testing of the entire web-based application before Exhibit 2. A special point I mentioned was that after completing the coding, it would be also important to consider potential code privacy and data security issues, such as how to encrypt the user water flow data from the sensor in a Text file.

Visual Evidence:



Statement on Learning:

This week's studio work has given me a deeper understanding of front-end code development, which also involves team collaboration, code version management, and control over the overall progress of the project. Although EJS is a JavaScript language that I have never worked with before, being responsible for the "Home" and "Setting" pages has given me an understanding of a new lightweight dynamic web page building language, rather than staying with Java and PHP. I have also experienced the efficiency of GitHub Desktop in team code development and version control.

In addition, my promotion of dynamic page development has allowed me to reserve time for the backend and database, which has made me realise the absolute importance of time management in leading a successful project. Only in this way can we fully understand and consider potential data privacy and security issues contained in the code. This is also a key point mentioned in every DECO course. Therefore, I have now developed the habit of always thinking about privacy and security when coding.

Design Computing Studio 2

Design Journal

Student Name: Xianglong Lin Student Number: 45791439 Team Name: H2OMG

Week: _12_

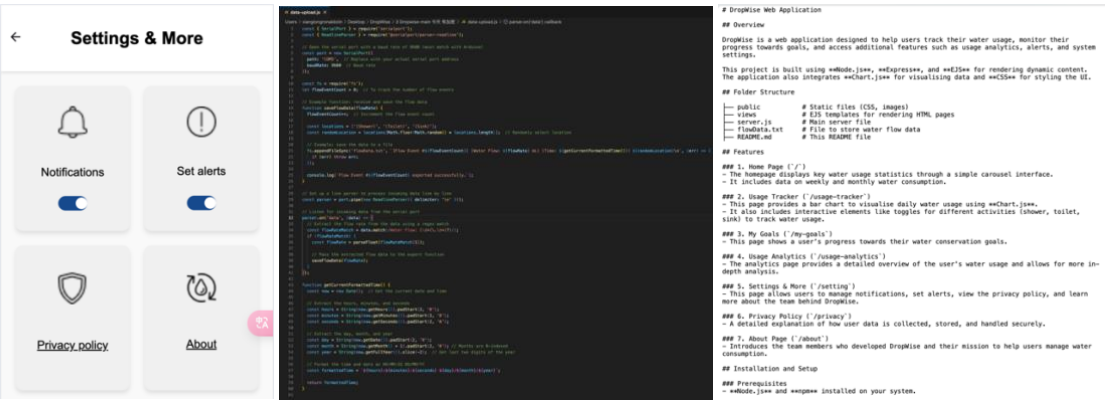
Title: Final Front-End Interface and Data Encryption Implementation in the Water Resource Protection System

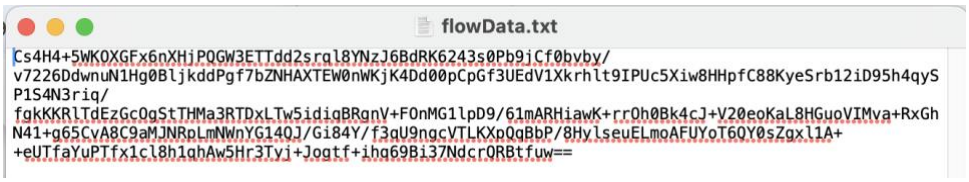
Description:

In the studio in the 12th week near the end of the course, I completed the last part of my front-end dynamic interface - the setting page, which allows users to turn on and off notifications and alerts and view related privacy policy and about details. Meanwhile, based on the data security consideration I proposed last week, I implemented the AES-256 encryption algorithm in the JsS code to encrypt the water flow data transmitted by the water flow sensor and store it in a local file.

Specifically, I used the crypto library in Node.js to generate a random encryption key and initialisation vector (IV) before saving the encrypted water flow data to a file. Furthermore, I added appropriate English comments to all the code and took the initiative to write a README document so that the mentor could understand our project's working principle and encryption process more quickly and better in Exhibit 2.

Visual Evidence:





Statement on Learning:

This week's studio not only enabled me to further enhance the security of the project by implementing data encryption, but also ensured the overall security of the system and the privacy protection of user data. I taught myself how to encrypt sensor data using the AES-256 algorithm, and I recognised the critical role of data encryption in preventing data leakage.

On the other hand, the writing of code comments and README documents has made me realise the importance of code readability and documentation for team collaboration and project presentation. Documentation not only helps team members better understand how the system works, but also makes it easier for mentors to mark and provide feedback in Exhibit 2.

Through systematic learning of DECO2850, I have deeply strengthened my multi-faceted skills required for developing secure, effective, and scalable systems, including but not limited to programming, data encryption, security considerations, and documentation work. This will also become my immediate strength in employment after near graduation.